

AP CALCULUS AB - UNIT 5

Guided Notes 5.1: Mean Value Theorem



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we verify continuity and differentiability hypotheses for the Mean Value Theorem?

- Verify continuity and differentiability hypotheses for the Mean Value Theorem.
- Find or guarantee points where instantaneous and average rates agree.
- Apply Rolle's Theorem as a special case.
- Distinguish existence conclusions from exact location and from the Intermediate Value Theorem.

Key Knowledge, Vocabulary, and Notation

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then some $c \in (a, b)$ satisfies $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Rolle's Theorem

If the MVT hypotheses hold and $f(a) = f(b)$, then some $c \in (a, b)$ has $f'(c) = 0$.

Geometric meaning

At some interior point, a tangent line is parallel to the secant through the endpoints.

Existence statement

The theorem guarantees at least one c but does not necessarily identify it or guarantee uniqueness.

Hypothesis failure

A corner, cusp, discontinuity, or vertical tangent can prevent use of the theorem.

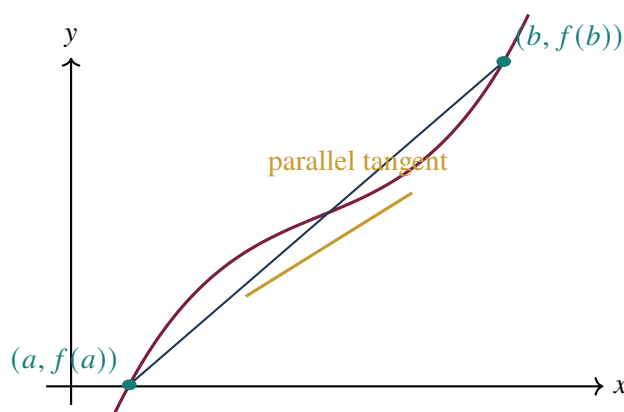
IVT distinction

IVT concerns attained function values; MVT concerns derivative values matching an average slope.

Concept Development and Strategy

1. State both hypotheses on their correct intervals before writing the conclusion.
2. Compute the endpoint secant slope, then solve $f'(c)$ equal to that slope and keep only solutions in (a, b) .
3. Do not use endpoint values alone to claim differentiability.
4. A theorem may be inapplicable even if a point satisfying the conclusion happens to exist.
5. In context, interpret the average and instantaneous rates with units.

Shared Representation / Data



Worked Example: Polynomial MVT

Verify the MVT for $f(x) = x^2 - 2x$ on $[0, 3]$ and find c .

Answer and Reasoning

A polynomial is continuous and differentiable. The secant slope is $(3 - 0)/3 = 1$. Since $f'(x) = 2x - 2$, solve $2c - 2 = 1$ to get $c = 3/2$.

Worked Example: Rolle's Theorem

Apply Rolle's Theorem to $f(x) = x^3 - 4x$ on $[-2, 2]$.

Answer and Reasoning

The polynomial is continuous/differentiable and $f(-2) = f(2) = 0$. Solve $f'(c) = 3c^2 - 4 = 0$: $c = \pm 2/\sqrt{3}$, both interior.

Worked Example: Hypothesis failure

Can the MVT be applied to $f(x) = |x|$ on $[-1, 1]$?

Answer and Reasoning

No. The function is continuous, but it is not differentiable at $x = 0$.

Worked Example: Context

A car travels 180 km in 2 hours with differentiable position. What does the MVT guarantee?

Answer and Reasoning

At some time during the trip, instantaneous velocity equals the average velocity 90 km/h.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. State the two MVT hypotheses.

2. Find the average slope of $f(x) = x^2$ on $[1, 4]$.

3. Apply the MVT to $f(x) = 1/x$ on $[1, 4]$ and find c .

4. Determine whether MVT applies to $f(x) = \sqrt{x}$ on $[0, 4]$.

5. Find all MVT points for $f(x) = x^3 - 3x$ on $[-2, 2]$.

6. Prove $x^5 + x + 1 = 0$ has at most one real root.

7. Use MVT to prove $|\sin b - \sin a| \leq |b - a|$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. MVT requires a function to be

- (A) continuous on (a, b) only (B) differentiable on $[a, b]$
(C) continuous on $[a, b]$ and differentiable on (a, b) (D) increasing on $[a, b]$

2. Rolle's Theorem is the MVT case in which

- (A) $f'(a) = f'(b)$ (B) $f(a) = f(b)$
(C) $a = b$ (D) f is linear

3. For $f(x) = \ln x$ on $[1, 5]$, the MVT point is closest to

- (A) 1.61 (B) 2.49
(C) 3.11 (D) 4.00

Common Errors and AP Precision

- A zero derivative or zero second derivative is a candidate condition, not by itself a complete extremum or inflection justification.
- State theorem hypotheses, include endpoints in absolute-extrema work, and verify every optimization candidate on the feasible domain.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

If $f'(x) > g'(x)$ for all x in (a, b) and $f(a) = g(a)$, prove $f(x) > g(x)$ for $x > a$.

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